Streaming for free: one weight vector for the filter and the smoother

Anonymous Author(s)
Affiliation
Address
email

Abstract

A two-stage sequence regressor that must run as a causal filter while data arrive and as a smoother once the window is complete leaks if the corrector is allowed to read the label during training. It learns to copy neighbours, and the two modes diverge at deployment. We train the corrector on the model's own pointwise draft only, and draw a causal or bidirectional mask per batch with equal probability. Two short propositions bound the gap (label leakage is a covariate shift bounded by the leaked fraction times the draft error; the mask draw puts both modes in one objective). The ordering of protocols is the same on oil temperature, air quality and ECG as on the aerodynamic case study where we found it. On ETTh1 a 4k-parameter model with one lag of the target and the covariates reaches 0.65 °C in both modes, where zero-shot Chronos, TimesFM and TinyTimeMixer given 512 hours of the target reach 0.75–0.81. The case study is a 5,285-parameter polar predictor that streams at 81 μs per 68-point window on one CPU thread.

1 Introduction

While a series is arriving the model must be a causal filter, using only the past. Once the window is complete it may use the whole context, as a smoother or a short-horizon simulator. One set of weights is what the device can store. Scheduled sampling [Bengio et al., 2015] lets a refinement stage read the true target with some probability; at deployment that channel is gone, and streaming and smoothing disagree.

We study a two-stage regressor: a pointwise draft $d(x)$ and a sequence corrector g_θ that reads neighbour drafts under a mask $m \in \{\mathrm{c}, \mathrm{b}\}$. Two training choices close the stream/smooth gap to $< 1\%$ for every model we trained. The corrector never sees the label, and each batch is causal or bidirectional with probability $\frac{1}{2}$. The same ordering of protocols holds on ETTh1 [Zhou et al., 2021], Beijing PM2.5 [Liang et al., 2015] and MIT-BIH [Moody and Mark, 2001]. Zero-shot time-series foundation models on the same ETTh1 hours take no covariates and run in one mode (0.75–0.81 °C vs 0.65 with one lag and the covariates, §3).

The airfoil polar is where we found the protocol. A small UAV wants C_l and C_d against α as it pitches, at the $(\mathrm{Re}, \mathrm{Ma})$ of the moment, on a processor shared with the flight loop. The window is about 75 steps, the context is shape plus flight condition, and the same weights have to filter while α sweeps and smooth once the polar is in. NeuralFoil [Sharpe, 2025] is a 13k-to-1.4M MLP family on XFOIL [Drela, 1989] sweeps; no member streams, predicts its own error, or takes Mach. FoilCORE is a 5,285-parameter model with a geometry code, a selective-scan draft, a two-block attention corrector and an error head (Figure 1). Transformer and Mamba correctors of the same size, trained the same way, match it. The protocol and the geometry code carry the accuracy.

Submitted to the FMTS Workshop at the 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Table 1: Test MAE of the same 4k-parameter draft→corrector as a stream and as a smoother, by protocol (mean of 3 seeds). Lower block: transformer corrector (4.5k).

Training protocol	ETTh1 oil temp. (°C)		Beijing PM2.5 (log)		MIT-BIH lead II (mV)	
	stream	smooth	stream	smooth	stream	smooth
label-leaking (sched. sampling)	5.52	5.27	0.590	0.502	0.300	0.289
draft-only, causal	5.19	5.30	0.489	0.515	0.229	0.226
draft-only, bidirectional	5.26	5.30	0.600	0.499	0.257	0.233
draft-only, dual mask (ours)	5.30	5.38	0.494	0.503	0.229	0.228
Same, with a transformer corrector in place of ours:						
label-leaking	6.03	5.69	0.732	0.582	0.297	0.295
draft-only, causal	4.62	4.67	0.487	0.581	0.227	0.229
draft-only, bidirectional	4.67	4.07	0.628	0.494	0.246	0.220
draft-only, dual mask	4.70	4.78	0.492	0.497	0.223	0.220

Related work. Exposure bias is old [Ranzato et al., 2016]; DAgger [Ross et al., 2011] queries an expert on the learner's states, Professor Forcing [Lamb et al., 2016] matches hidden states adversarially. Dual-mode ASR [Yu et al., 2021, Zhang et al., 2020] and UniLM/UL2 [Dong et al., 2019, Tay et al., 2023] already train under mixed masks. What we add is the draft-only rule for a corrector reading a neighbour series of its *own target* (the leakage channel of Proposition 1, absent in ASR) and evidence it survives at 5k parameters and on three public datasets. Foundation forecasters [Ansari et al., 2024, Das et al., 2024, Ekambaram et al., 2024] appear zero-shot in §3.

2 Draft-only dual-mask training

Let x be a window's conditions, $y \in \mathbb{R}^T$ its targets, $\hat{y} = d(x)$ the pointwise draft, and $g_\theta(x, \tilde{y}, m)$ the corrector reading neighbour values $\tilde{y}$ under mask m. Deployment risk in mode m is $R_m(\theta) = \mathbb{E}\,\ell(g_\theta(x, \hat{y}, m), y)$. Scheduled sampling with leak fraction q trains on $\tilde{y}_t = y_t$ w.p. q, else $\hat{y}_t$, giving $\tilde{R}^q_m(\theta)$.

Proposition 1 (label leakage is a covariate shift; its cost is bounded by q times the draft error). If $\tilde{y} \mapsto \ell(g_\theta(x, \tilde{y}, m), y)$ is L-Lipschitz in $\|\cdot\|_1$, then $|R_m(\theta) - \tilde{R}^q_m(\theta)| \le L\,q\,\mathbb{E}\|y - \hat{y}\|_1$. *Proof:* $|R_m - \tilde{R}^q_m| \le \mathbb{E}|\ell(g_\theta(x, \hat{y}, m), y) - \ell(g_\theta(x, \tilde{y}, m), y)| \le L\,\mathbb{E}\sum_t |\tilde{y}_t - \hat{y}_t|$, and $\tilde{y}_t - \hat{y}_t$ equals $y_t - \hat{y}_t$ with probability q and 0 otherwise. The training-risk minimizer is a deployment-risk minimizer only when $q = 0$ or the draft is exact; when the draft is imperfect (exactly when a corrector is worth having) training rewards reading the leaked label. Draft-only training sets $q = 0$. L is finite for the correctors we train (linear maps, GELU, LayerNorm, softmax, a linear scan; ℓ_1 is 1-Lipschitz) but weight-dependent, so the bound is qualitative. Table 1 shows the predicted ordering.

Proposition 2 (one set of weights for both modes). With $m \sim \mathrm{Bernoulli}(\frac{1}{2})$ the objective is $\bar{R} = \frac{1}{2}(R_\mathrm{c} + R_\mathrm{b})$, so its minimizer $\bar{\theta}$ satisfies $R_m(\bar{\theta}) \le R_\mathrm{c}(\theta) + R_\mathrm{b}(\theta)$ for every θ. A specialist has no guarantee in the other mode: bidirectional specialists in Table 1 stream 10–27% worse than they smooth. The stronger two-specialist bound needs realizability and fails in general; it approximately holds here, with one disclosed exception in §3.

The leaking baseline is Bengio et al. [2015]'s anneal, not a fixed mix: q starts at 1 and steps down to 0. The gap survives the anneal (0.129/0.0082 smoothing vs. 0.137/0.0107 streaming on the same 3.4k airfoil trunk).

3 Does the recipe transfer?

We strip every airfoil-specific part, keep a 4k-parameter pointwise-MLP draft, two-block attention corrector and residual head, and train under four protocols on three public conditional streaming-regression tasks: ETTh1 (loads and calendar → oil temperature, 96-h windows, held-out four-week blocks), Beijing PM2.5 (weather → log PM2.5, 72-h windows, split by year) and MIT-BIH (one ECG lead → lead MLII at 90 Hz, 128 samples, split by record).

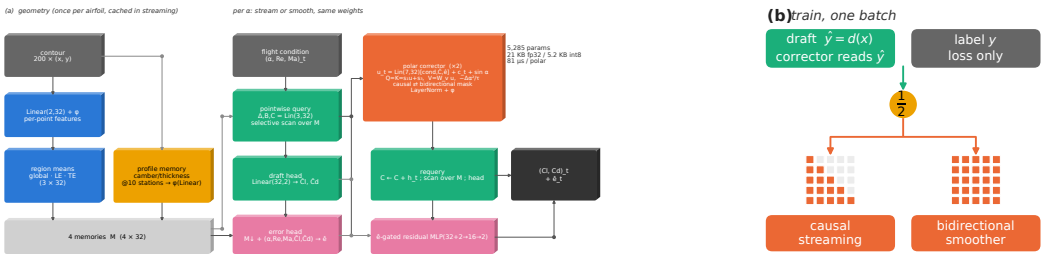


Figure 1: (a) FoilCORE. (b) The corrector reads only $\hat{y} = d(x)$; each batch is causal or bidirectional with probability $\frac{1}{2}$.

Table 2: Held-out-geometry test (34,043 polars). Causal AR unless marked. NeuralFoil on a 1,024-polar subset. Latency: T=68, one CPU thread.

Model	Params	C_l	C_d (stall)	smooth. C_l/C_d	MACs/pt	latency	framework
FoilCORE-5k (ours)	5,285	0.046	0.0036 (0.0060)	0.046 / 0.0036	12.8k	81 μs	C++ kernel
FoilCORE-2k (d=16)	2,373	0.057	0.0044 (0.0073)	0.058 / 0.0043	6.1k	61 μs	C++ kernel
FoilCORE-3k (plain)	3,434	0.076	0.0052 (0.0089)	0.075 / 0.0052	11.8k	67 μs	C++ kernel
FoilCORE-0.5k (d=8)	482	0.100	0.0083 (0.0129)	0.100 / 0.0084	2.6k	38 μs	C++ kernel
3k, label-leaking	3,434	0.137	0.0107 (0.0180)	0.129 / 0.0082	11.8k	68 μs	C++ kernel
Same front-end and protocol, standard backbone as corrector:							
Transformer	5,822	0.048	0.0037 (0.0061)	0.047 / 0.0036	8.1k	166 μs	C++ kernel
TSMixer	11,780	0.061	0.0046 (0.0075)	0.061 / 0.0045	2.4k*	50 μs	C++ kernel
Mamba (S6)	5,300	0.049	0.0038 (0.0063)	0.049 / 0.0038	3.7k*	122 μs	C++ kernel
NeuralFoil xxsmall	13,302	0.092	0.0128 (0.0256)	–	26k	0.57 ms	NumPy
NeuralFoil xxxlarge	1,428,166	0.082	0.0115 (0.0239)	–	2.9M	4.63 ms	NumPy

The label-leaking protocol is the worst streamer everywhere (5.52 vs. 5.19–5.30 °C, 0.590 vs. 0.489–0.494, 0.300 vs. 0.229 mV), the only one with a systematic stream-vs-smooth gap (+5%, +18%, +4%), and on PM2.5 and ECG worse than every draft-only variant even as a smoother. The dual-mask model matches the streaming specialist as a stream and the smoothing specialist as a smoother, within seed spread, from one set of weights. A transformer corrector repeats the picture (lower block); a Mamba corrector does too (released package). Three notes. On ETTh1 the draft-only specialists have no gap either (learned attention is local). On ECG the corrector barely improves the pointwise draft: reconstructing lead MLII from the simultaneous second lead is almost a pointwise map, so the protocol effect is only that leaking hurts. With the transformer on ETTh1 the bidirectional specialist is the better smoother (4.07 vs. 4.78): the one cell where future context pays is also where one weight vector does not attain both optima, Proposition 2's realizability caveat in the data.

Foundation forecasters on the same hours. Zero-shot Chronos-Bolt (8.7M and 48M), TinyTimeMixer-r2 (0.8M) and TimesFM-2.5 (231M) forecast ETTh1 oil temperature one hour ahead from 512 hours of its own past at 0.79, 0.78, 0.81 and 0.75 °C (last value: 0.80). Given one lag of the target plus the current covariates, the 4k model reaches 0.65 °C as a stream and as a smoother (a covariate-matched ridge also reaches 0.65 with the lag, 5.8 without it vs. our 5.3), at 0.3 ms per window on one CPU thread against 3–54 ms (TimesFM: 97 ms on a GPU). They take no covariates and run in one mode. Scale did not remove a gap that the protocol creates.

4 Case study: a 5,285-parameter polar predictor

Task: contour $\{(x_i, y_i)\}_{i=1}^{200}$ and $\{(\alpha_t, \mathrm{Re}_t, \mathrm{Ma}_t)\}_{t=1}^{T}$ ($T \approx 75$) $\rightarrow (C_l, C_d)_t$. Split 80/10/10 by near-duplicate geometry cluster on 5,392 airfoils (XFOIL, ncrit = 9; BigFoil), so test is unseen shapes (539 geometries, 34k polars). Metric: pooled per-point MAE, causal first.

FoilCORE-5k reaches 0.046 C_l / 0.0036 C_d streaming (3 seeds 0.048±0.004 / 0.0037±0.0002); causal and bidirectional agree to 1%. NeuralFoil xxxlarge remains 1.2×/1.4× better in incompressible mid-Re pre-stall; FoilCORE wins overall because it takes Mach and is 2.7–4× better in stall. With the same front-end and protocol, a transformer (5,822 p) and Mamba (5,300 p) match it; TSMixer

(11,780 p) does not. A fused C++ kernel runs a polar in 81 μs (0.55 mJ package-level). Mean-pooled geometry codes collapse to 3–7 principal dimensions; a parameter-free camber/thickness profile as a fourth memory is the largest architectural gain (dropping it: $0.047 \rightarrow 0.410$ C_l). Analytic lift and drag priors (518 parameters) stack on top of the protocol (next paragraphs). Trained on 10% of the geometries the same model still reaches 0.081/0.0065. The error head's C_d estimate has Spearman 0.71 with the draft residual and rises 4.8$\times$ on a non-airfoil cloud.

We split by geometry cluster and score the full test set (CSV prefixes are low-Re biased). Per-channel SSM correctors lose to attention; a stall router never fired; depth-3 was worse.

Width d=32. Each contour point: shared $\mathrm{Linear}(2, d)$ and a per-channel path nonlinearity ϕ (Chebyshev series plus a Gaussian bump; pretrained as an autoencoder). Three memories are masked means over the whole contour, leading edge ($x \leq 0.25$) and trailing edge ($x \geq 0.75$). The fourth is camber and thickness at ten cosine chord stations (fixed 20-vector, no parameters) mapped by $\mathrm{Linear}(20, d)$ and ϕ. For every $(\alpha, \mathrm{Re}, \mathrm{Ma})$ a selective scan [Gu and Dao, 2024, Gu et al., 2022] over the four memories yields a draft $(\hat{C}_l, \hat{C}_d)$; a 16-unit MLP predicts $|\mathrm{draft\ error}|$. Tokens $[\alpha, \mathrm{Re}, \mathrm{Ma}, \hat{C}_l, \hat{C}_d, \hat{e}]$ plus context and $\mathrm{sinusoid}(\alpha)$ pass two residual-product-attention blocks (Q=K=$s_1 u + s_3$, V=$W_v u$, scores $-\Delta\alpha^2/\tau$, causal or full mask). Loss: MAE on final $+\,0.25\times$ draft MAE (z-scored, equal C_l/C_d) $+\,0.1\times$ error-head L1; AdamW 3×10^{-4}, batch 1024, early stopping on causal validation loss. Training: 40 min per 100 epochs on an RTX A4500.

Three analytic levers add 518 parameters: a potential-flow lift base $2\pi \sin(\alpha - \alpha_0)/\sqrt{1 - \mathrm{Ma}^2}$ with a learned stall taper; a fifth memory over six contour scalars plus four boundary-layer scale features; a gated quadratic drag polar. On a second BigFoil pool they are additive over four seeds with complete separation: causal C_l 0.0506$\pm$0.0026 $\rightarrow$ 0.0416$\pm$0.0014. Seeded with one measured point, the first causal step beats the physics-free twin's full smoothing pass. They do not transfer zero-shot across pools (normalization uses training-pool statistics); replication retrains per pool.

5 Limitations

Labels are XFOIL at ncrit = 9. The transfer study tests protocol ordering with one small model per dataset, not each dataset's literature; the foundation-model rows are zero-shot on a different information set. Ablation rows are single runs; the main model is three seeds.

Train the refinement stage on the model's own outputs, in both masks, and the deployment mode stops being a compromise.

References

Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, et al. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024.

Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.

Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning (ICML)*, 2024.

Li Dong, Nan Yang, Wenhui Wang, Furu Wei, Xiaodong Liu, Yu Wang, Jianfeng Gao, Ming Zhou, and Hsiao-Wuen Hon. Unified language model pre-training for natural language understanding and generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Mark Drela. XFOIL: An analysis and design system for low reynolds number airfoils. *Low Reynolds Number Aerodynamics*, pages 1–12, 1989.

Vijay Ekambaram, Arindam Jati, Pankaj Dayama, Sumanta Mukherjee, Nam H. Nguyen, Wesley M. Gifford, Chandra Reddy, and Jayant Kalagnanam. Tiny time mixers (TTMs): Fast pre-trained models for enhanced zero/few-shot forecasting of multivariate time series. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *First Conference on Language Modeling (COLM)*, 2024.

Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. In *International Conference on Learning Representations (ICLR)*, 2022.

Alex M. Lamb, Anirudh Goyal, Ying Zhang, Saizheng Zhang, Aaron C. Courville, and Yoshua Bengio. Professor forcing: A new algorithm for training recurrent networks. In *Advances in Neural Information Processing*

Systems (NeurIPS), 2016.

Xuan Liang, Tao Zou, Bin Guo, Shuo Li, Haozhe Zhang, Shuyi Zhang, Hui Huang, and Song Xi Chen. Assessing Beijing's PM2.5 pollution: severity, weather impact, APEC and winter heating. *Proceedings of the Royal Society A*, 471(2182):20150257, 2015.

George B. Moody and Roger G. Mark. The impact of the MIT-BIH arrhythmia database. *IEEE Engineering in Medicine and Biology Magazine*, 20(3):45–50, 2001.

Marc'Aurelio Ranzato, Sumit Chopra, Michael Auli, and Wojciech Zaremba. Sequence level training with recurrent neural networks. *International Conference on Learning Representations (ICLR)*, 2016.

Stéphane Ross, Geoffrey J. Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2011.

Peter Sharpe. NeuralFoil: An airfoil aerodynamics analysis tool using physics-informed machine learning. *arXiv:2503.16323*, 2025.

Yi Tay, Mostafa Dehghani, Vinh Q. Tran, Xavier Garcia, Jason Wei, Xuezhi Wang, Hyung Won Chung, Siamak Shakeri, Dara Bahri, Tal Schuster, Huaixiu Steven Zheng, Denny Zhou, Neil Houlsby, and Donald Metzler. UL2: Unifying language learning paradigms. In *International Conference on Learning Representations (ICLR)*, 2023.

Jiahui Yu, Wei Han, Anmol Gulati, Chung-Cheng Chiu, Bo Li, Tara N. Sainath, Yonghui Wu, and Ruoming Pang. Dual-mode ASR: Unify and improve streaming ASR with full-context modeling. In *International Conference on Learning Representations (ICLR)*, 2021.

Binbin Zhang, Di Wu, Zhuoyuan Yao, Xiong Wang, Fan Yu, Chao Yang, Liyong Guo, Yaguang Hu, Lei Xie, and Xin Lei. Unified streaming and non-streaming two-pass end-to-end model for speech recognition. *arXiv preprint arXiv:2012.05481*, 2020.

Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *AAAI Conference on Artificial Intelligence*, 2021.